Dyadic descaling and delayed Pell sign recovery

Two explicit reductions for Diophantine expressions of Con(ZFC)

Prepared for Vladimir Reshetnikov
Research, implementation, and separate proof review with ChatGPT

23 October 2026

Abstract

We construct two explicit polynomial equations whose unsolvability is equivalent to Con(ZFC), using precisely stated calibration theorems already present in ProveIt. The first equation has ten strictly positive unknowns, exact total degree 706, and a division-free circuit with 133 binary arithmetic operations. Its immediate parent has degree 816 and 124 operations. The essential change is to separate four rational square classes by distinct powers of two. This removes a strong-Pell scale factor that had been needed for real-size separation. Three centers and a quadratic transport majorant then determine the lower exact degree.

The second equation is a quartic in 780 unrestricted integers, defined both by 434 explicit quadratic residuals and by all $9,628$ collected monomials. It improves the 888-integer parent in the latest direct certificate package. Its largest absolute coefficient is $2,445,790,974,704$, unchanged from that parent. This route uses the direct C222 proof-certificate calibration rather than K215 or Jones's universal equation. The reduction removes 36 nonnegativity guards by proving their consequences in a strict order: elementary signs, the Pell frame, indexed powers, and finally signs of power quotients.

The new algebraic and number-theoretic reductions are proved here; the two source calibrations are inherited, not silently replaced by experiments. Literal data, exact rebuilding programs, source hashes, audit receipts, and an independent AI-assisted proof review accompany the article. We do not claim global minimality, a new universal-variable record, or an end-to-end proof-assistant certificate.

Contents

1 The target and the two results **3**
1.1 What changes mathematically . . . 3

2 Inherited calibration and precise proof boundaries **4**
2.1 The complete fixed coefficient tuple . . . 4
2.2 The native positive source interface . . . 4
2.3 The independent direct source interface . . . 5

3 A degree-706 equation in ten positive integers **5**
3.1 The polynomial and its fixed coefficients . . . 6
3.2 The norm identity . . . 7
3.3 Square classes separated by powers of two . . . 8
3.4 Soundness: extracting transport and the canonical data . . . 10
3.5 Completeness and the positive sum coordinate . . . 13
3.6 A symbolic proof of the exact degree . . . 14
3.7 Circuit form, verification and limitations . . . 15

4 A direct quartic in 780 unrestricted integers **16**
4.1 Exact object and inherited source . . . 16
4.2 Guarded signs and a staged inference certificate . . . 17
4.3 The frame bounds do not require power-quotient signs . . . 17
4.4 Index certification precedes power recovery . . . 18
4.5 The quotient sign follows from the exact power . . . 18
4.6 Literal interface and absence of circularity . . . 19
4.7 Integer projection and the resulting theorem . . . 19

5 Verification and reproducible artifacts **20**
5.1 Ten-positive checks . . . 21
5.2 Direct-quartic checks . . . 21
5.3 Files and exact commands . . . 21

6 Limitations, interpretation, and further research **22**
6.1 What has and has not been achieved . . . 22
6.2 Questions suggested by the reductions . . . 22

A Domain conventions and conversion **23**

B Attribution and source access **23**

1 The target and the two results

Fix the usual recursive presentation of ZFC used by the repository's calibration. Write $\mathrm{Ref}_{\mathsf{ZFC}}(p)$ for its primitive-recursive predicate that p is a finite syntactic refutation, and abbreviate

$$\mathrm{Con}(\mathsf{ZFC}) \;\equiv\; \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathsf{ZFC}}(p).$$

The desired assertion for a polynomial P and an explicitly specified domain D is

$$\mathsf{ZFC} \vdash (\forall \boldsymbol{x} \in D^v\ P(\boldsymbol{x}) \neq 0) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}). \tag{1.1}$$

The useful way to establish this is to prove, internally by ordinary arithmetizable mathematics, that a zero exists exactly when a finite refutation exists. There is no assertion here that ZFC proves its own consistency, or that a numerical search could establish unsolvability.

The word "simplest" requires a choice of cost. Unknown count, degree, arithmetic circuit length, number of monomials, coefficient height, compressed coefficient storage, and proof dependencies are different measures. A sparse quartic with hundreds of unknowns can have much smaller coefficients than a compact formula in ten unknowns. This work therefore states concrete improvements and their costs rather than an unproved absolute optimum.

Construction	Unknowns	Domain	Degree	Further size measure
Immediate scaled-norm parent	10	$\mathbb{Z}_{>0}$	816	124 operations
Dyadic construction here	10	$\mathbb{Z}_{>0}$	**706**	133 operations
Direct parent	888	$\mathbb{Z}$	4	10,156 monomials
Delayed-sign construction here	**780**	$\mathbb{Z}$	4	9,628 monomials

Table 1: Exact comparisons to the two pinned immediate parents. Domains and coefficient bindings are kept distinct.

The first change reduces degree by $110/816 \approx 13.48\%$ and adds nine arithmetic operations. It does not dominate its parent in every cost. The second removes $108/888 \approx 12.16\%$ of the integer unknowns while preserving degree and coefficient height. It is a reduction within the direct proof-certificate architecture, not a claim that no smaller integer quartic occurs elsewhere in the repository.

All comparisons use the fixed repository snapshot `ec80841c07e91be8956ce6ed4af9a9c0b782d3d9`. The reviewed collection already contains nine-positive-variable and ten-integer equations, and earlier native-calibration quartics have far fewer than 780 integer unknowns. The latter use a different coefficient regime. The direct 407-natural-variable parent is also retained unchanged by our new integer-domain conversion. These comparisons prevent a domain change or an obsolete baseline from being mistaken for a record.

1.1 What changes mathematically

The ten-positive-variable parent makes several real radicals grow at different rates, so an irrational square class cannot disappear from a signed sum. That requirement inflates its strong Pell ordinate. Here we arrange instead that each radicand is a unit at the prime 2, then weight the four roots by $1, 2, 4, 8$. Within any common rational square class, the weighted coefficients have distinct powers of two. The least power cannot cancel. The same rational-root conclusion follows without size ordering, and one scale factor can be deleted.

The direct construction addresses a different source of growth. An integer version of a natural-number certificate often attaches three new variables to each sign requirement. Some sign requirements become theorems after the arithmetic meaning of the certificate is established. We prove those meanings before inferring the remaining signs, and record the exact dependency order. Removing 36 such requirements saves $3 \cdot 36 = 108$ integer variables.

Neither the use of valuations nor the elementary quotient recurrence is claimed as a new general principle. The contributions are their concrete applications, the compatible proof ordering, the complete fixed polynomials, and the quantified improvements at the pinned source. General work on universal pairs illustrates why degree and variable count must be tracked separately [5]; Sun's ten-integer theorem concerns a different, universal reduction [6]. Neither new endpoint here invokes Sun's ten-integer theorem.

2 Inherited calibration and precise proof boundaries

2.1 The complete fixed coefficient tuple

For the first construction the fixed numerals are

$$\beta = 2^{590069} - 1, \qquad K_0, \qquad M_C, \qquad M_F.$$

The last three are literal integers supplied by the unchanged compact44 coefficient files. Each file is a gzip-compressed little-endian binary integer. Their complete compressed and uncompressed SHA-256 bindings appear in the circuit JSON and its builder. The designation M_F means the *already shifted* high mask. There is no additional shift. These fixed numerals satisfy

$$K_0 > 0, \qquad 0 < M_C < \beta < M_F < 2\beta. \tag{2.1}$$

They are part of the polynomial's definition, not existential parameters, an unspecified enumeration index, or a truth-dependent choice. The four compressed payloads together occupy about nine kilobytes; their uncompressed representation has 295,388 bytes.

Numeral	Supplied file	Bit length
β	`beta.bin.gz`	590,069
K_0	`K.bin.gz`	592,884
M_C	`MC.bin.gz`	590,069
M_F	`MF_source.bin.gz`	590,070

Table 2: The whole compact44 binding. No ports from another tuple are mixed in.

2.2 The native positive source interface

Here is the exact inherited theorem used for the ten-positive equation. It is Theorem `thm:source` of package 61, including the canonical promises immediately following it [2]. Define, for positive source coordinates

$$J, F, \alpha, z, f, h, i, T_{\mathrm{old}}, s, w, \tau, \eta, \zeta, y, \gamma,$$

the registers

$$q = \beta J + 1, \qquad C = q - F - \alpha, \qquad X = wq, \qquad Y = sq^3,$$
$$E = XY, \qquad k = \eta + \zeta, \qquad c = kY + \eta, \qquad a = E + Y,$$
$$H_0 = 4a + 3, \qquad \Delta_0 = (a+2)^2 - 1, \qquad \ell = EkY, \qquad I_0 = ic^2.$$

Set

$$R = ((q-1)(q-F)+\alpha)(q^2-1)+(M_C+qM_F)J,$$
$$D = X + ac + H_0\gamma, \qquad V_{\text{old}} = c(T_{\text{old}}f - 1) - Rf^2.$$

The source equations are

$$\begin{aligned} \tau^2 - \ell(\ell+k) &= 1, & D^2 - \Delta_0 c^2 &= 1, \\ f^2 - \Delta_0 I_0^2 &= 1, & \Delta_0(f^2-1)(V_{\text{old}}^2 - y^2) + y^2 &= 1, \\ k - hE - R &= 1, & (K_0+w)C + (q-F) - z(q-1) &= 1. \end{aligned} \tag{2.2}$$

Theorem 2.1 (Inherited calibrated positive source)**.** *For the entire tuple just specified, a positive solution of* (2.2) *yields a finite syntactic refutation of ZFC. Conversely, a finite refutation yields positive solutions with*

$$\begin{aligned} &q = B^N, \quad B = \beta + 1, \quad N > 1 \text{ odd}, \quad J = 1 + B + \cdots + B^{N-1}, \\ &R \equiv 3 \pmod 4, \qquad X = 2^R, \qquad c = \psi_{a+2}(R), \\ &k = 2\psi_{2XY^2+1}((R+1)/2), \qquad \tau = \chi_{2XY^2+1}((R+1)/2). \end{aligned}$$

The two implications are ordinary mathematics formalizable in ZFC.

Here $\chi_A(n) + \psi_A(n)\sqrt{A^2-1} = (A+\sqrt{A^2-1})^n$. The source theorem composes the native compiler, its compact44 adapter, and the literal blank-K215/refutation calibration. We inherit that composition. The source is retained and pinned in the package; this article does not re-prove every compiler and machine lemma. The immediate degree-816 proof [1] gives the precise Pell rank, index, parity, ratio, and auxiliary-completeness interfaces used below, and is included unchanged for inspection.

2.3 The independent direct source interface

The second route starts with the exact 690-natural-variable quadratic residual system in the pinned incoming package *ConZFC_407_888_Quartics* [4]. Its natural zero set is calibrated to finite ZFC refutations through the direct C222/Horn certificate, beta coding, binomial arithmetic, and shared Pell frames. Its existing positive-to-integer conversion has 888 unknowns. Our transformation changes the sign certificates and then performs exact monic eliminations. It preserves the source system's natural zero set. The original archive, including the source explanation and provenance, is retained byte for byte.

Thus the logical dependency has two independently specified forms:

$$N_{706} = 0 \text{ in } \mathbb{Z}_{>0}^{10} \longleftrightarrow \text{native positive source zero} \longleftrightarrow \exists p\ \text{Ref}_{\mathsf{ZFC}}(p),$$
$$P_{780} = 0 \text{ in } \mathbb{Z}^{780} \longleftrightarrow \text{direct natural source zero} \longleftrightarrow \exists p\ \text{Ref}_{\mathsf{ZFC}}(p).$$

Negating each chain gives (1.1). The new reductions do not upgrade the inherited chain to an exported object-language ZFC proof or a complete Lean/Rocq formalization.

3 A degree-706 equation in ten positive integers

The first construction preserves the ten positive unknowns of the scaled-norm candidate in the pinned repository, while reducing its exact total degree from 816 to 706. Its division-free circuit uses 133

binary arithmetic operations, compared with 124 for that parent. This is a degree–operation tradeoff, rather than a claim that one measure of simplicity captures every consideration. The coefficient tuple and its calibration remain those described in Section 2. The immediate mathematical parent is [1]; the source interface and the complete fixed coefficient binding are [2, 3].

The main change is to make all four rational radicands units at the prime 2. Distinct powers of 2 then separate their rational square classes. This removes a size-order condition on the strong Pell root and permits one factor to be removed from its ordinate. Three centers in the carrier and a quadratic transport majorant account for the remaining reduction. We prove the new algebra and positivity arguments explicitly. The inherited compiler/calibration and the stated Pell index lemmas retain precisely the status recorded in Section 2.

3.1 The polynomial and its fixed coefficients

Write the four fixed integer coefficients as

$$\beta = 2^{590069} - 1, \qquad K_0, \qquad M_C, \qquad M_F.$$

The complete binary payloads are supplied with the circuit and authenticated by their compressed and uncompressed SHA-256 digests. Their verified bounds are

$$K_0 > 0, \qquad 0 < M_C < \beta < M_F < 2\beta. \tag{3.1}$$

Here M_F is already the shifted high mask; it must not be shifted a second time. None of these four constants is a quantified unknown.

The ten unknowns are

$$(j, F, \alpha, i, T, w, Y, n, \eta, \zeta) \in \mathbb{Z}_{>0}^{10}, \qquad \mathbb{Z}_{>0} = \{1, 2, 3, \ldots\}. \tag{3.2}$$

Every register in the following definition is an integer polynomial in these unknowns and the four fixed constants. Define the outer registers by

$$\begin{aligned} J &= 2j + 1, & q &= \beta J + 1, & C &= q - F - \alpha, \\ X &= wq, & a &= Y(X + 1), & k &= \eta + \zeta, \\ c &= kY + \eta, & \mu &= q^2 + 1, & \lambda &= 2\mu. \end{aligned} \tag{3.3}$$

The scaled main and first Pell parameters are

$$\begin{aligned} A &= \lambda(a + 2) + 1, & H &= 4A - 5, & \Delta &= A^2 - 1, \\ a_- &= A - 2, & V_0 &= X\lambda^2 Y^2, & D_\tau &= V_0(V_0 + 1)k^2 + 1. \end{aligned} \tag{3.4}$$

The packed index, its excess and the transport numerator are

$$\begin{aligned} G &= (q - 1)(q - F) + \alpha, \\ R &= G(q^2 - 1) + (M_C + qM_F)J, \\ e &= k - R - 1, & B_t &= (K_0 + w)C - F. \end{aligned} \tag{3.5}$$

For the strong and auxiliary Pell registers put

$$\begin{aligned} I &= ic^2, & D_f &= 1 + \Delta I^2, & Q &= \Delta^2 I^2, & \delta &= Q - 1, \\ T' &= T + \alpha, & b &= cT' - R, & V &= D_f b - c, \\ D_y &= QV^2 - 1, & D_m &= \Delta c^2 + 1. \end{aligned} \tag{3.6}$$

In particular, I has no factor μ other than any accidental divisibility of its value. Define the centers and transport weight

$$\begin{aligned} C_f &= AI, & C_\tau &= V_0 k, \\ L &= (k+F+1)(T'+6), & W &= 1+(q-1)L, \end{aligned} \tag{3.7}$$

and the denominator-clearing registers

$$\begin{aligned} P_1 &= q^3(q-1), & M &= P_1\mu^2 H, \\ Z &= P_1\mu^2\big[H(n+V+2C_f+4C_\tau)+8(X+a_-c)\big] \\ &\quad + H\big[\mu^2((q-1)Y-q^3WB_t)-P_1 e\big]. \end{aligned} \tag{3.8}$$

For the final norm use the five polynomial arguments

$$\begin{aligned} s &= \delta Z^2, & u_1 &= 16\delta M^2 D_\tau, & u_2 &= 4\delta M^2 D_f, \\ u_3 &= M^2 D_y, & u_4 &= 64\delta(P_1\mu^2)^2 D_m. \end{aligned} \tag{3.9}$$

Here is a compact fixed norm polynomial in five formal arguments. Its letters a, b, c, d are local formal arguments, independent of the similarly named registers above:

$$\begin{aligned} \sigma_1 &= a+b+c, & \sigma_2 &= ab+(a+b)c, & \sigma_3 &= abc, \\ U &= s+d, & P &= U-\sigma_1, & E &= P^2+4(sd-\sigma_2), \\ G_* &= E^2+16(sdP^2-4\sigma_3 U), & O_* &= EP-16\sigma_3, & & \\ \mathcal{F}_{16}(s;a,b,c,d) &= G_*^2-64sdO_*^2. & & & & \end{aligned} \tag{3.10}$$

The equation delivered here is

$$\boxed{N=\mathcal{F}_{16}(s;u_1,u_2,u_3,u_4)=0, \quad (j,F,\alpha,i,T,w,Y,n,\eta,\zeta)\in\mathbb{Z}_{>0}^{10}.} \tag{3.11}$$

The named registers are abbreviations for substitution, not additional unknowns. Equations (3.3)–(3.10) are consequently a finite, exact specification of one integer-coefficient polynomial.

Theorem 3.1 (Ten-positive equation)**.** *With the fixed coefficient binding and source calibration of Section 2, the polynomial N has a zero in $\mathbb{Z}_{>0}^{10}$ if and only if ZFC has a finite syntactic refutation. The reduction is ordinary mathematics formalizable in ZFC. Thus the source calibration gives*

$$\mathrm{ZFC}\vdash\big[(\forall\mathbf{x}\in\mathbb{Z}_{>0}^{10})\,N(\mathbf{x})\neq 0\big]\longleftrightarrow\mathrm{Con}(\mathrm{ZFC}).$$

The exact total degree of N is 706*. The supplied literal circuit has* 133 *binary operations:* 79 *multiplications,* 33 *additions and* 21 *subtractions, including operations with small numerals.*

3.2 The norm identity

We include the conjugate-product calculation because it specifies exactly what the single polynomial equation asserts.

Lemma 3.2 (Sixteen conjugates)**.** *In the polynomial ring over $\mathbb{Z}$,*

$$\mathcal{F}_{16}(x^2;r_1^2,r_2^2,r_3^2,r_4^2)=\prod_{\varepsilon\in\{1,-1\}^4}(x-\varepsilon_1 r_1-\varepsilon_2 r_2-\varepsilon_3 r_3-\varepsilon_4 r_4). \tag{3.12}$$

Moreover, $\mathcal{F}_{16}$ is homogeneous of degree 8 *in its five arguments and has leading coefficient* 1 *in s.*

Proof. Put $a = r_1^2$, $b = r_2^2$, $c = r_3^2$ and $d = r_4^2$. Pairing the signs of r_1 and r_2, then the sign of r_3, gives the three-root identity

$$\prod_{\varepsilon_1,\varepsilon_2,\varepsilon_3} (y - \varepsilon_1 r_1 - \varepsilon_2 r_2 - \varepsilon_3 r_3) = \left[(y^2 - \sigma_1)^2 - 4\sigma_2\right]^2 - 64\sigma_3 y^2.$$

For example the first pairing is $(y^2 - a - b)^2 - 4ab$; substituting $y - r_3$ and $y + r_3$ and multiplying yields the displayed expression.

Now fix the sign ε of r_4 and set $y = x - \varepsilon r_4$, $s = x^2$ and $r = 2xr_4$. Then $y^2 = U - \varepsilon r$ and $r^2 = 4sd$. Consequently

$$(y^2 - \sigma_1)^2 - 4\sigma_2 = E - 2\varepsilon r P.$$

The three-root product becomes

$$(E - 2\varepsilon r P)^2 - 64\sigma_3(U - \varepsilon r) = G_* - 4\varepsilon r O_*.$$

Multiplication over the two values of ε gives $G_*^2 - 16r^2 O_*^2 = G_*^2 - 64sdO_*^2$, as claimed. The product identity also shows the homogeneity and the coefficient of s^8. □

For later use define positive real roots, whenever the positive input is fixed, by

$$\tau = \sqrt{D_\tau}, \qquad f = \sqrt{D_f}, \qquad y = \sqrt{D_y/\delta}, \qquad D = \sqrt{D_m}, \qquad d = D/H. \tag{3.13}$$

Their existence for *every* positive input is checked below, before any equation or transport sign is assumed. Lemma 3.2 then yields

$$N = \delta^8 M^{16} \prod_{\varepsilon \in \{1,-1\}^4} \left(\frac{Z}{M} - 4\varepsilon_1\tau - 2\varepsilon_2 f - \varepsilon_3 y - 8\varepsilon_4 d \right). \tag{3.14}$$

Direct division of the definition of Z gives the explanatory identity

$$\frac{Z}{M} = n + V + 2C_f + 4C_\tau + \frac{8(X + a_- c)}{H} + \frac{Y}{q^3} - \frac{WB_t}{q-1} - \frac{e}{\mu^2}. \tag{3.15}$$

Only the polynomial definition (3.8), not division, is used in evaluating N.

3.3 Square classes separated by powers of two

Lemma 3.3 (Dyadic separation)**.** *Let $r_1, \dots, r_m$ be positive real numbers with rational squares and $v_2(r_i^2) = 0$. Let $h_1, \dots, h_m$ be distinct integers. If, for some signs ϵ_i, the sum $\sum_i \epsilon_i 2^{h_i} r_i$ is rational, then all the r_i are rational.*

Proof. The extension $K = \mathbb{Q}(r_1, \dots, r_m)$ is a finite multiquadratic Galois extension. Its group sends each r_i to $\pm r_i$, defining a character χ_i. Two roots have the same character exactly when their ratio is rational; a character is trivial exactly when its root is rational. If some nontrivial character χ occurs, apply its projection

$$\frac{1}{|\operatorname{Gal}(K/\mathbb{Q})|} \sum_{g \in \operatorname{Gal}(K/\mathbb{Q})} \chi(g) g$$

to the rational relation. It yields

$$\sum_{\chi_i = \chi} \epsilon_i 2^{h_i} r_i = 0.$$

Divide by any root in this nonempty group. The ratios are rational and have 2-adic valuation zero, because their squares have valuation zero. The resulting rational summands have pairwise distinct valuations h_i. The term of least valuation cannot cancel. Equivalently, clear the odd denominators and divide by the least power of two; one summand is odd and all the others are even. This contradiction excludes every nontrivial character. □

The lemma uses valuations of rational ratios only. It does not require an extension of the rational 2-adic valuation to arbitrary irrational real numbers, and it imposes no relative-size assumption on the roots.

Lemma 3.4 (All-input positivity and units)**.** *For every input in $\mathbb{Z}_{>0}^{10}$, δ, D_y, H are positive and the four radicands in (3.13) are positive rational numbers of 2-adic valuation zero.*

Proof. The number β is odd, as is J, so q is even and $q \geq 16$. Thus μ is odd, λ is even, A is odd, Δ is even, V_0 is even and H is odd. Also $k \geq 2$, $c \geq 3$, $I \geq c^2$ and $\Delta > \mu^2 > 1$. In particular $Q > 1$ and $\delta > 0$.

If the integer $b = cT' - R$ is zero, then $V = -c$ and $|V| \geq 3$. If $b \neq 0$, then

$$|V| = |D_f b - c| \geq D_f - c > 1,$$

since $D_f \geq 1 + \Delta c^4$. Hence $D_y = QV^2 - 1 > 0$ in both cases. Every numerator and denominator in

$$D_\tau, \qquad D_f, \qquad \frac{D_y}{\delta}, \qquad \frac{D_m}{H^2}$$

is odd: $D_\tau = 1 + V_0(V_0+1)k^2$, $D_f = 1 + \Delta I^2$, $Q = \Delta^2 I^2$ is even, and $D_m = 1 + \Delta c^2$. All four rational numbers are therefore units at 2. □

Lemma 3.5 (Adjacent-coefficient descent)**.** *Let $Q > 1$ and V be integers. If $y \in \mathbb{Q}$ satisfies*

$$QV^2 - (Q-1)y^2 = 1,$$

then y is an integer.

Proof. Replace V, y by their absolute values. Neither can be zero. Write $y = u/m$ in lowest terms, with $m > 0$. The equation implies $m^2 \mid Q - 1$. When $V = 1$, it gives $y = 1$. For $V > 1$ define

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV.$$

The first quantity is an integer because $m \mid Q - 1$. The identity $(2Q-1)^2 - 4Q(Q-1) = 1$ gives

$$Q(V')^2 - (Q-1)(y')^2 = 1.$$

Furthermore

$$(Q-1)(y^2 - V^2) = V^2 - 1 > 0$$

shows $y > V$, whereas

$$(QV)^2 - ((Q-1)y)^2 = QV^2 + Q - 1 > 0$$

shows $QV > (Q-1)y$. These inequalities imply $-V < V' < V$. The new equation excludes $V' = 0$. Since $2Q - 1 \equiv 1 \pmod{m}$, the reduced denominator of y' is still m. Induction on the positive integer $|V|$, using $|V'| < V$, reaches $V = 1$. Thus $m = 1$. □

If $N = 0$, equation (3.14) gives a rational signed sum of the roots with weights $(1, 2, 4, 8)$ in the order (y, f, τ, d). Lemmas 3.4 and 3.3 make all four roots rational. The roots τ, f, D have integer squares and hence are integers. Lemma 3.5 applied to $QV^2 - (Q-1)y^2 = 1$ makes y an integer as well. This step is valid even when V is still negative: the descent can use $|V|$.

3.4 Soundness: extracting transport and the canonical data

Assume $N = 0$ at a positive input, and choose signs from a vanishing factor of (3.14). The first congruence is obtained before transport positivity. Since q is even,

$$\gcd(\mu^2, q^3(q-1)) = 1, \qquad H = 4\lambda(a+2) - 1 \equiv -1 \pmod{\mu}.$$

Thus μ^2 is coprime to $q^3(q-1)H$. Integer-valuedness of the signed roots in (3.15) gives

$$\frac{8(X + a_- c - \varepsilon_4 D)}{H} + \frac{Y}{q^3} - \frac{WB_t}{q-1} - \frac{e}{\mu^2} \in \mathbb{Z}.$$

Clearing the other denominators therefore gives $\mu^2 \mid e$. Put

$$\rho = \frac{e}{\mu^2} \in \mathbb{Z}, \qquad g = \frac{\varepsilon_4 D - X - a_- c}{H}, \qquad s_0 = \frac{Y}{q^3} > 0, \qquad t = \frac{B_t}{q-1}. \tag{3.16}$$

At this stage g, s_0, t need not be integers. The carrier identity is

$$n = 4(\varepsilon_1 \tau - C_\tau) + 2(\varepsilon_2 f - C_f) + (\varepsilon_3 y - V) + 8g - s_0 + Wt + \rho. \tag{3.17}$$

Use the standard Pell notation

$$\chi_B(r) + \psi_B(r)\sqrt{B^2 - 1} = (B + \sqrt{B^2 - 1})^r.$$

The integer roots give, for positive indices,

$$\begin{aligned} \tau &= \chi_P(n_0), & k &= 2\psi_P(n_0), & P &= 2V_0 + 1, \\ D &= \chi_A(p), & c &= \psi_A(p), & & \\ f &= \chi_A(m), & I &= \psi_A(m). & & \end{aligned} \tag{3.18}$$

For the first equation the fundamental solution of $x^2 - V_0(V_0+1)k^2 = 1$ is $(2V_0+1, 2)$: $k = 1$ would require a square strictly between V_0^2 and $(V_0+1)^2$. If $n_0 = 1$, then $k = 2$, $\eta = \zeta = 1$ and $c = 2Y + 1$. But $\psi_A(1) = 1$ and $\psi_A(p) \geq 2A > 2Y + 1$ for $p \geq 2$. Consequently $n_0 \geq 2$, and

$$c > k \geq 4P > 8X\lambda^2 Y^2 > 8q^5. \tag{3.19}$$

The mask bound (3.1) gives

$$0 < (M_C + qM_F)J < 2q^2.$$

If $F \geq q$, then $G \leq \alpha$; if $F < q$, then $G < (q-1)^2 + \alpha$. In either case

$$R < q^4 + (\alpha + 2)q^2. \tag{3.20}$$

Together with (3.19), this proves $c(T + \alpha) > R$. Hence $b \geq 1$, $V = D_f b - c > 1$ and $y > V$. No conclusion about B_t has yet been used.

Lemma 3.6 (Quadratic transport bound)**.** *Every positive zero of N satisfies $B_t \geq 0$.*

Proof. Before any sign restriction on R, the definition of R gives

$$|R| < q^4 + Fq^3 + (\alpha + 2)q^2. \tag{3.21}$$

As $q \ge 16$ and $\mu^2 > q^4$, it follows that

$$\frac{|R|}{\mu^2} < F + \alpha + 1, \qquad |\rho| \le \frac{k + 1 + |R|}{\mu^2} < k + F + \alpha + 1. \tag{3.22}$$

The root centers satisfy the one-sided estimates

$$\tau - C_\tau < k, \qquad f - C_f < 0, \qquad 8g < k. \tag{3.23}$$

For the first inequality, square the comparison with $(V_0+1)k$. For the second, $D_f = A^2I^2 - I^2 + 1 < A^2I^2$, since $I > 1$. For the third, $D < Ac$ implies $g < 2c/H$; now $c < 2kY$ and $H > 32Y$. These upper bounds apply to the corresponding signed roots in (3.17) as well.

Put $u = y - V$. Since $y > V > 1$,

$$u = \frac{V^2 - 1}{(Q-1)(y+V)} < \frac{V}{2(Q-1)} < \frac{D_f b}{2(Q-1)} < \frac{b}{\Delta}.$$

The last inequality is equivalent to $\Delta D_f = Q + \Delta < 2Q - 2$, which holds because $Q = \Delta^2 I^2 > \Delta + 2$. Also $\Delta > \mu^2$ and $c < 2kY < k\Delta$. Thus

$$u < kT' + F + \alpha + 1. \tag{3.24}$$

If the integer B_t were negative, it would be at most -1, and $Wt \le -W/(q-1) < -L$. Substituting (3.22)–(3.24) into (3.17), and using $\varepsilon_3 y - V \le y - V$ and $s_0 > 0$, yields

$$n < kT' + 6k + 2F + 2\alpha + 2 - L < 0.$$

Indeed

$$\begin{aligned} L - (kT' + 6k + 2F + 2\alpha + 2) &= (F+1)T' + 4F + 4 - 2\alpha \\ &= (F+1)T + (F-1)\alpha + 4F + 4 > 0. \end{aligned}$$

This contradicts the stipulated $n \ge 1$. □

Now $(K_0 + w)C \ge F > 0$, so $C \ge 1$. The source outer estimates give

$$3q < R < q^4 - 2. \tag{3.25}$$

For completeness, write $G = (q-1)C + q\alpha$ and use $C + \alpha = q - F \le q - 1$. Its lower bound and the positive mask term imply the left inequality; $G \le q^2 - q$ and the mask bound below $2q^2$ imply the right one.

We record how the inherited index arguments apply; this is also a check that deleting the factor μ from I changes no hypothesis. The first Pell congruence is

$$k \equiv 2n_0 \pmod{4V_0}, \qquad \mu^2 \mid 4V_0.$$

Together with $\mu^2 \mid k - R - 1$ and (3.25), it gives $2n_0 \ge R + 1$. Equations (3.19) and (3.25) give $c > 2R$; the main Pell equation gives $c > 2p$. The rank lemma requires $c^2 \mid I$, which holds exactly as written in (3.6). It gives $c \mid m$ and $m > 2p$. With $S = \Delta I$ the auxiliary data satisfy

$$S^2 = \Delta(f^2 - 1) = Q, \qquad c \mid S, \qquad V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c}.$$

The source index lemma therefore gives $p = R$. Since $P > A$, monotonicity of Pell ordinates gives $n_0 < R$. The congruence $2n_0 \equiv R + 1 \pmod{\mu^2}$, together with $R < q^4 < \mu^2$, forces $2n_0 = R + 1$. The fixed-minus parity lemma then implies $R \equiv 3 \pmod 4$.

Set $r = (R-1)/2$. The parent ratio lemma is valid for every integer scaling $\lambda \geq 1$ and hence for $\lambda = 2\mu$. In its notation,

$$\xi = \frac{(X+1)^{2r}}{X^r}, \qquad \xi < \frac{\psi_A(2r+1)}{\psi_P(r+1)} < \xi\left(1 + \frac{3}{a}\right)^{2r}. \tag{3.26}$$

The defining interval $kY < c < k(Y+1)$ gives

$$Y > \frac{X^r}{3}, \qquad a > \frac{X^{r+1}}{3}, \qquad r \geq 24. \tag{3.27}$$

Before the exponent has been determined, clearing the three remaining fractions gives

$$H \mid 8q^3(q-1)(X + a_- c - \varepsilon_4 D).$$

Because H is odd, the factor 8 cancels. The main projection identities are

$$D - a_- c \equiv 2^R \pmod H, \qquad D + a_- c \equiv 2^{-R} \pmod H.$$

The parent Archimedean estimate, using (3.27), is

$$q^4, 2^{R+1}X < X^{r+1} < 3a < H. \tag{3.28}$$

If $\varepsilon_4 = -1$, multiplication by 2^R would make $q^3(q-1)(2^R X + 1)$ a positive multiple of H smaller than H. This is impossible. If $\varepsilon_4 = +1$, the same estimate bounds $|q^3(q-1)(X - 2^R)|$ strictly below H, so divisibility gives

$$\varepsilon_4 = +1, \qquad X = 2^R. \tag{3.29}$$

In particular g is an integer. The remaining partial fractions now give $q^3 \mid Y$ and $q - 1 \mid B_t$.

The ratio estimate (3.26) and the half-binomial lemma yield

$$Y = \frac{1}{2}\sum_{h=0}^{r} \binom{2r}{r+h} X^h. \tag{3.30}$$

To see the uniqueness mechanism, write

$$\xi = M_0 + \theta, \qquad M_0 = \sum_{h=0}^{r} \binom{2r}{r+h} X^h, \qquad 0 < \theta < \tfrac{1}{4}.$$

Here M_0 is even because $X = 2^{2r+1}$ and the central binomial coefficient is even. The quantitative ratio estimate puts c/k in $(M_0/2, M_0/2 + 1)$, while its defining interval is $(Y, Y+1)$. These two integer parts must agree. The explicit estimates and source reconstruction are the ones in [1, Sections 6.17–6.18]; only their arbitrary positive scaling parameter has changed.

Finally keep J, F, α, w, Y and set $s = Y/q^3$, $z = B_t/(q-1) + 1$. Reconstruct fresh *unscaled* Pell witnesses with main parameter $a + 2$ and first parameter $2XY^2 + 1$, using (3.29), $R \equiv 3 \pmod 4$ and (3.30). This is the source reconstruction of [1, Section 6.18]: it requires only the recovered canonical data and supplies all fifteen positive witnesses of the calibrated source system. Theorem `thm:source` of [2] gives a finite syntactic refutation of ZFC. This proves soundness relative to the calibrated interface.

3.5 Completeness and the positive sum coordinate

Suppose ZFC has a finite syntactic refutation. The calibrated source provides a canonical positive solution with

$$q = B^N \text{ for odd } N > 1, \quad R \equiv 3 \pmod 4, \quad X = 2^R, \quad C > 0,$$

and Y equal to (3.30). Its odd J gives the positive integer $j = (J-1)/2$. The parent ratio argument, valid for arbitrary positive integer scaling, constructs main and first Pell values for $\lambda = 2\mu$:

$$c = \psi_A(R), \quad D = \chi_A(R), \quad k = 2\psi_P((R+1)/2), \quad \tau = \chi_P((R+1)/2).$$

Set $\eta = c - kY$, $\zeta = k - \eta$; both are positive. The quantities

$$\rho = (k - R - 1)/\mu^2, \qquad g = (D - X - a_- c)/H, \qquad t = B_t/(q-1)$$

are integers with $\rho, g \geq 1$ and $t \geq 0$, as in the parent completeness proof.

For the strong Pell pair use the smaller index

$$m = Rc, \qquad I = \psi_A(m), \qquad f = \chi_A(m). \tag{3.31}$$

The multiplication formula and the standard Pell congruence give

$$\psi_A(Rc) = c\psi_D(c), \qquad \psi_D(c) \equiv cD^{c-1} \pmod{D^2 - 1}.$$

Since $c^2 \mid D^2 - 1$, this implies $c \mid \psi_D(c)$ and therefore $c^2 \mid I$. Thus $i = I/c^2$ is a positive integer. No divisibility by μ is needed.

Let $S = \Delta I$ and $Q = S^2$. The explicit solution in the auxiliary completeness lemma has

$$\begin{aligned} V^* &= \frac{\chi_S(R)}{S} = Q_v(S^2), & y &= \psi_S(R), & v &= (R-1)/2, \\ T^* &= \frac{V^* + c + Rf^2}{cf^2}, & Q(V^*)^2 - (Q-1)y^2 &= 1. \end{aligned} \tag{3.32}$$

The inherited lemma proves that T^* is an integer. The associated odd Pell polynomial obeys

$$Q_v(z) \geq (4z - 3)^v \quad (z \geq 1).$$

Since $v \geq 2$ here,

$$V^* \geq (4Q - 3)^v \geq (3Q)^v \geq 9Q^2. \tag{3.33}$$

As $c < S$ and $f^2 = D_f < Q$, it follows that

$$T^* > \frac{V^*}{S^3} \geq 9S > \alpha.$$

The last comparison uses the canonical $C > 0$, hence $\alpha < q < S$. Define $T = T^* - \alpha \geq 1$; then $T' = T^*$ and $V = V^*$ as required.

There is one new completeness issue: the center $C_f = AI$ lies above f, so its contribution to n is negative. The auxiliary gap has ample room to absorb it, with an explicit bound.

Lemma 3.7 (Canonical auxiliary gap)**.** *For the witnesses just constructed, $u = y - V$ satisfies*

$$u > 2I + Y + 1.$$

Proof. Since $V \geq 2$, $Q \geq 3$ and $V < y < 2V$,

$$u = \frac{V^2 - 1}{(Q-1)(y+V)} > \frac{V}{4Q} \geq \frac{9S^2}{4} > 3S + 1.$$

For the first strict inequality one may use $V^2 - 1 \geq \frac{3}{4}V^2$ and $(Q-1)(y+V) < 3QV$. The second is (3.33). Finally $Y < c < I < S$, so $3S + 1 > 2I + Y + 1$. □

Define the integer

$$n = Wt + \rho + 8g + 4(\tau - C_\tau) + 2(f - C_f) + u - Y/q^3. \tag{3.34}$$

The terms Wt, ρ, g are nonnegative or positive, and $\tau > C_\tau$. Moreover $f > (A-1)I$, so $f - C_f > -I$, and $Y/q^3 \leq Y$. Lemma 3.7 therefore gives $n > 0$, hence $n \in \mathbb{Z}_{>0}$. Substitution into (3.15) yields

$$Z/M = 4\tau + 2f + y + 8d.$$

The all-positive sign factor in (3.14) vanishes, proving $N = 0$. This establishes completeness and the semantic part of Theorem 3.1.

3.6 A symbolic proof of the exact degree

The literal circuit gives the conservative syntactic degree 832. That estimate ignores cancellations caused by the three centers. We prove the exact total degree using a formal parameter, so no numerical specialization is needed for the upper bound.

Replace every unknown x by tx, keeping the direction variables algebraically independent, and work over their rational function field K. Degree below means degree in t at infinity. In $K((t^{-1}))$ each relevant square root has a branch whose leading term is its center: this follows by the binomial expansion of $(1+h)^{1/2}$ when h has negative degree. For y, choose the branch with leading term V. These formal branches need not be positive real branches; the product over all signs is independent of that choice.

The register and error degrees are

Quantity	Degree
μ, λ	2
A, H, Δ	$5, 5, 10$
I, Q, δ	$5, 30, 30$
D_f, V, M	$20, 24, 13$
C_f, C_τ	$10, 9$
$\tau - C_\tau$	1
$f - C_f$	0
$y - V$	-6
d	2
L, W, B_t	$2, 3, 2$
$Z/M - V - 2C_f - 4C_\tau$	4

For example,

$$f^2 - C_f^2 = 1 - I^2$$

has degree 10, while $f + C_f$ has degree 10. Similarly

$$(y - V)(y + V) = \frac{V^2 - 1}{Q - 1}$$

gives degree $48 - 30 - 24 = -6$ for $y - V$. The degree-4 leading term of the remaining carrier is $-L_2(B_t)_2$, coming from $-WB_t/(q-1)$; its other terms have degree at most 2.

The sixteen factors in (3.14) have the following nonzero leading terms:

Signs	Count	Degree	Leading term
$\varepsilon_3 = -1$	8	24	$2V_{24}$
$\varepsilon_3 = 1$, $\varepsilon_2 = -1$	4	10	$4(C_f)_{10}$
$\varepsilon_3 = \varepsilon_2 = 1$, $\varepsilon_1 = -1$	2	9	$8(C_\tau)_9$
$\varepsilon_3 = \varepsilon_2 = \varepsilon_1 = 1$	2	4	$-L_2(B_t)_2$

Subscripts here denote leading homogeneous forms. Their product has exact degree

$$8 \cdot 24 + 4 \cdot 10 + 2 \cdot 9 + 2 \cdot 4 = 258.$$

The prefactor $\delta^8 M^{16}$ has degree $8 \cdot 30 + 16 \cdot 13 = 448$. Thus

$$\deg N = 258 + 448 = 706. \tag{3.35}$$

For an explicit check that the leading form is nonzero, put

$$\begin{aligned}
q_0 &= 2\beta j, & k_0 &= \eta + \zeta, & c_0 &= k_0 Y,\\
\lambda_0 &= 2q_0^2, & A_0 &= \lambda_0 w q_0 Y, & \Delta_0 &= A_0^2,\\
I_0 &= i c_0^2, & Q_0 &= \Delta_0^2 I_0^2, & &\\
V_* &= -(\Delta_0 I_0^2) q_0^3 (q_0 - F), & & & &\\
C_{f,0} &= A_0 I_0, & C_{\tau,0} &= w q_0 \lambda_0^2 Y^2 k_0, & &\\
M_0 &= 4 q_0^8 A_0, & L_0 &= (k_0 + F)(T + \alpha), & B_{t,0} &= w(q_0 - F - \alpha).
\end{aligned}$$

The degree-706 component is

$$2^{22} Q_0^8 M_0^{16} V_*^8 C_{f,0}^4 C_{\tau,0}^2 (L_0 B_{t,0})^2. \tag{3.36}$$

Every factor is a nonzero polynomial. More concretely, on the diagonal where all ten unknowns equal t, put $q_0 = 2\beta$ as a fixed integer. The coefficient of t^{706} is

$$9 \cdot 2^{202} q_0^{366} (q_0 - 1)^8 (q_0 - 2)^2 > 0. \tag{3.37}$$

This explicit coefficient supplies a lower-degree witness for the actual fixed coefficient tuple. The three other coefficient ports occur only in lower-degree terms.

3.7 Circuit form, verification and limitations

The proof uses W as a convenient abbreviation. The literal circuit does not materialize it: it distributes

$$\frac{WB_t}{q-1} = LB_t + \frac{B_t}{q-1}. \tag{3.38}$$

Thus the circuit subtracts LB_t directly from $n + V + 2C_f + 4C_\tau$, and uses B_t alone in the residual fractional numerator. This exact identity saves one binary operation and leaves the polynomial unchanged. The final count is $79 + 33 + 21 = 133$; all rows are live and every unknown and coefficient is used.

The supplied build program checks the four compressed and raw coefficient hashes, their bit lengths and bounds, the circuit topology, liveness and operation counts, and deterministic regeneration of the JSON circuit. The audit evaluates the displayed formulas independently at modular points and computes complete degree-bounded univariate restrictions modulo two primes. These restrictions have degree 706 and the leading coefficient (3.36); the diagonal restrictions also match (3.37). Additional exact finite cases check the parity and majorant premises and the lower bound on the canonical auxiliary gap. These are implementation checks supporting the proof, rather than a substitute for it.

The reduction is a measurable improvement in degree over the pinned parent, with a modest increase in circuit length. It does not claim the least possible degree, the fewest possible unknowns, or the shortest possible equation. It retains the inherited K215/native compiler route and its stated formalization boundary. For a formulation over nonnegative integers, replacing each unknown x by $x + 1$ preserves the exact degree and adds at most ten additions to the literal evaluator.

4 A direct quartic in 780 unrestricted integers

The second construction reduces the direct C_{222} quartic of [4] from 888 to 780 integer unknowns. It keeps degree four, reduces the collected support from 10,156 to 9,628 monomials, and leaves the maximum absolute expanded coefficient unchanged at $2\,445\,790\,974\,704$. Its new ingredient is an order of proof: the signs of the private power quotients are recovered *after* the power graphs have been shown correct. This removes independent sign guards without weakening the final arithmetic specification.

4.1 Exact object and inherited source

Let $r_0, \ldots, r_{433}$ be the literal quadratic rows in this component's `equations/equation.json`, using its listed order of 780 variables. Define the fixed integer polynomial

$$P_{780}(x_0, \ldots, x_{779}) = \sum_{j=0}^{433} r_j(x)^2. \tag{4.1}$$

All 780 variables range over $\mathbb{Z}$. The complete collected polynomial is also supplied as `expanded_polynomial.json`; this is an ordinary polynomial with literal integer coefficients and fixed exponents. The coefficient of x_0^4 is 3, so its degree is exactly four.

The pinned immediate source is the archive `ConZFC_407_888_Quartics_2026-10-24.zip` at ProveIt commit `ec80841c07e91be8956ce6ed4af9a9c0b782d3d9`. It is retained byte for byte, with its original notices; the separate `SOURCE_PINS.json` records the archive digest and the exact members used. Write $S(y) = 0$ for its 606 quadratic source rows on $y \in \mathbb{N}^{690}$. The inherited arithmetic and logical calibration is

$$(\exists y \in \mathbb{N}^{690} : S(y) = 0) \iff C_{222}(1) \iff \exists p \ \mathrm{Prf}_{\mathrm{ZFC}}(p, \bot). \tag{4.2}$$

The source report identifies the inherited calibration theorems

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation
```

Their calibration snapshot is

`08200eb10fde2db408c69dffd840d6920a569b1f`.

Here we prove the new existence equivalence with S; the proof of (4.2) is inherited from that explicitly identified source.

4.2 Guarded signs and a staged inference certificate

For a selected source coordinate y_v, three private integer unknowns enforce the quadratic guard

$$y_v = a_v^2 + b_v^2 + c_v(c_v + 1). \tag{4.3}$$

This forces $y_v \geq 0$ for all integers a_v, b_v, c_v. Conversely, every nonnegative integer has such a representation: apply Legendre's three-square theorem to $4y_v + 1$, whose residue modulo four forces one of the three squares to be odd and the other two even, then subtract one and divide by four. The parent used 167 such guards. We use 131.

Ordinary sign propagation is performed on the unchanged source rows. An oriented row is factored exactly as $a(\widehat{y})y_v - b(\widehat{y}) = 0$, with neither a nor b involving y_v. Previously proved lower bounds $a \geq 1$ and $b \geq 0$ imply $y_v \geq 0$; if $b \geq 1$, integrality implies $y_v \geq 1$. The checker establishes these bounds from sums of nonnegative monomials and exact affine squares. Odd powers need an established nonnegative factor; even powers need no sign premise. A monomial whose factors are all at least one contributes its coefficient to the constant lower bound. This last rule can certify a polynomial with a negative constant. Every premise must have been established earlier in the certificate.

Starting from the 131 guards, these elementary inferences prove 623 of the 690 source coordinates nonnegative. They also prove that every power base and every exponent is at least one, and prove all sign premises for the frame below. In particular, a factorial output used as a power base receives its strict sign from literal arithmetic rows; the proof does not first appeal to factorial correctness.

4.3 The frame bounds do not require power-quotient signs

There are 23 power calls (n, k, m), nine distinct bases, and 47 distinct raw atoms occurring as a base, exponent, or proposed output. Denote that set of atoms by $\mathcal{A}$. The exact shared frame is

$$W = 2 + \sum_{a \in \mathcal{A}} a, \qquad Z = z + 1, \qquad A^2 - W^3(W+2)Z^2 = 1. \tag{4.4}$$

At this point all atoms in $\mathcal{A}$ are known nonnegative, $n, k \geq 1$, $A \geq 0$ and $Z \geq 1$. Consequently $W \geq 2$, $1 \leq n, k \leq W$, $0 \leq m \leq W$, and $A > W^2$.

For $a \geq 2$, define the Pell coordinates by

$$\begin{aligned} X_0(a) &= 1, & Y_0(a) &= 0, \\ X_{j+1}(a) &= aX_j(a) + (a^2-1)Y_j(a), & Y_{j+1}(a) &= X_j(a) + aY_j(a). \end{aligned} \tag{4.5}$$

The norm in (4.4) is also $A^2 - ((W+1)^2 - 1)(WZ)^2 = 1$. Pell-solution classification therefore gives $A = X_j(W+1)$ and $WZ = Y_j(W+1)$ for some $j > 0$. Reducing (4.5) modulo W gives $X_j(W+1) \equiv 1$ and $Y_j(W+1) \equiv j \pmod{W}$. Thus $W \mid j$ and $j \geq W$, whence

$$A \geq (W+1)^j \geq (W+1)^W > W^W \geq n^k. \tag{4.6}$$

For each distinct base the literal system defines

$$T_n = 2An - n^2 - 1, \qquad D = A^2 - 1. \tag{4.7}$$

Their signs have not been independently guarded. Since $1 \le n \le W < A$, the function $2An - n^2 - 1$ is increasing on this integer interval. Therefore

$$T_n \ge 2A - 2 > W^W \ge n^k, \qquad T_n > W \ge m, \qquad D \ge 1. \tag{4.8}$$

We have recovered the nine modulus signs and the discriminant sign using only the frame and earlier elementary facts. Elementary propagation from these facts raises the number of nonnegative source coordinates to 644. No private power quotient was used in this step.

4.4 Index certification precedes power recovery

Fifteen seed exponents have individual witnesses $X_i, Y_i, S_i, R_i, e_i, s_i, r_i$, with a shared U, V, B and auxiliary quotients b, v, q. The exact equations are

$$\begin{aligned} Y_i &= k_i + e_i, & X_i^2 - (A^2-1)Y_i^2 &= 1, \\ S_i^2 - (B^2-1)R_i^2 &= 1, & S_i &= X_i + Us_i, \\ R_i &= k_i + 4Y_i r_i, & U^2 - (A^2-1)V^2 &= 1, \\ B &= A + Ub, & V &= \left(\prod_{i=1}^{15} Y_i^2\right)(v+1), \\ B &= 1 + 4V(q+1). \end{aligned} \tag{4.9}$$

Every displayed witness has already been proved nonnegative, and $k_i \ge 1$. Hence $Y_i \ge 1$, $V \ge 1$, $B \ge 5$, $Y_i^2 \mid V$, and $4Y_i \mid (B-1)$. For each i these equations satisfy the hypotheses of the inherited Matiyasevich Pell-index criterion `PellRepresentation.criterion_iff_pell`. Thus

$$X_i = X_{k_i}(A), \qquad Y_i = Y_{k_i}(A). \tag{4.10}$$

The three remaining requested pairs are derived by shifts $s = 1$ or $s = 2$, using the exact polynomial identities

$$X_{k+s} = X_k X_s + DY_k Y_s, \qquad Y_{k+s} = X_k Y_s + Y_k X_s, \tag{4.11}$$

where $(X_1, Y_1) = (A, 1)$ and $(X_2, Y_2) = (2D+1, 2A)$. Their gate identities and exponent shifts are checked literally. All required Pell pairs are therefore identified before using the sign of any power quotient.

4.5 The quotient sign follows from the exact power

Lemma 4.1 (Delayed power-quotient sign)**.** *Suppose $A \ge 2$, $n, k \ge 1$, $T = 2An - n^2 - 1 > 0$, $0 \le m < T$ and $n^k < T$. Suppose further that $X = X_k(A)$, $Y = Y_k(A)$, and that an integer h satisfies*

$$X + nY = AY + m + Th. \tag{4.12}$$

Then $m = n^k$ and

$$h = \sum_{j=0}^{k-1} n^{k-1-j} Y_j(A) \ge 0. \tag{4.13}$$

Proof. Put $H_j = X_j(A) - (A-n)Y_j(A)$. Direct substitution into (4.5) gives

$$H_0 = 1, \qquad H_{j+1} = nH_j + TY_j(A).$$

Induction yields

$$H_k = n^k + T\sum_{j=0}^{k-1} n^{k-1-j}Y_j(A). \tag{4.14}$$

Equations (4.12) and (4.14) show $m \equiv n^k \pmod{T}$, independently of the sign of h. Both numbers lie in $[0, T)$, so $m = n^k$. Substitution and cancellation of $T \neq 0$ then give (4.13). All its summands are nonnegative. □

The exact recurrence already occurs in the inherited positive-power construction. The new use is to apply its soundness argument with h unrestricted and then recover its sign. The preceding frame and index analysis verifies every hypothesis of Lemma 4.1 for all 23 calls. We obtain 23 quotient signs; a final elementary pass obtains the remaining 23 product signs. All 690 source coordinates are now nonnegative.

4.6 Literal interface and absence of circularity

The semantic interface is checked against the immutable raw gate ledger, which contains 1,301 coordinates and 1,217 rows. Source elimination replaced 611 of these rows by zero identities; the other 606 become exactly the literal rows of S. The independent checker reconstructs every omitted raw coordinate by its monic gate definition and verifies all these polynomial identities. This reconstruction is valid at integer zeros, before any nonnegativity assumption on the restored coordinates.

The interface file then binds (4.4), (4.7), every instance of (4.9), the three addition formulas, and all 23 equations (4.12) to exact raw rows and atoms. Some named atoms are related by arithmetic gates—for example, both t and $t+2$ occur in the distinct-atom sum. Identity comparison preserves those gate equations; it does not incorrectly declare such atoms algebraically independent.

The actual inference order, with checked counts, is

Stage	Source coordinates known nonnegative
131 private guards and elementary propagation	623
frame bounds and further elementary propagation	644
index identification, power recovery, final propagation	690.

All index premises are checked before invoking the power lemma. Power outputs enter W as already nonnegative proposed values; no claim that they are correct is used to derive the frame bounds. This establishes the sign argument in an ordered form.

4.7 Integer projection and the resulting theorem

Let G_{131} be the 606 source rows together with the 131 guards (4.3), on 1,083 unrestricted integer coordinates. The sign proof shows that every integer zero of G_{131} projects to a natural zero of S. Conversely any natural zero of S extends by 131 independent three-square representations. Therefore

$$\exists y \in \mathbb{N}^{690}\ S(y) = 0 \quad \longleftrightarrow \quad \exists z \in \mathbb{Z}^{1083}\ G_{131}(z) = 0. \tag{4.15}$$

Only after this conversion do we perform signed coordinate elimination. A row $z_v - f = 0$, or its negative, with f an integer polynomial not involving z_v, permits exact substitution everywhere.

Each reduced integer zero restores uniquely by $z_v = f$; positivity and divisibility conditions are unnecessary. The independent checker expands every substitution and checks that the remaining rows are quadratic. There are 303 eliminations: 188 affine and 115 quadratic. No unconstrained coordinates are discarded. Reverse traversal restores the entire guarded solution uniquely, leaving exactly 780 coordinates and 434 residuals. Thus

$$\exists z \in \mathbb{Z}^{1083}\ G_{131}(z) = 0 \quad \longleftrightarrow \quad \exists x \in \mathbb{Z}^{780}\ P_{780}(x) = 0. \tag{4.16}$$

Uniqueness concerns the monic restoration; the initial three-square extensions need not be unique.

Theorem 4.2. *Using the inherited calibration* (4.2)*, the explicit polynomial* (4.1) *satisfies*

$$\mathrm{ZFC} \vdash (\nexists x \in \mathbb{Z}^{780} : P_{780}(x) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

Proof. Compose (4.2), (4.15), and (4.16), using that an integer sum of squares is zero exactly when every summand is zero. Negate the resulting existence equivalence. The new reasoning is internalizable in ZFC using integer arithmetic, natural induction, standard Pell theory, and the three-square theorem; it assumes no consistency statement. □

The 36 guards removed from the immediate parent are the 23 private power quotients, nine moduli T_n, the discriminant D, and three coordinates handled by the stronger positive-factor rule: `endpoint.ut`, `factorial0.output`, and `crt.nonnegative_quotient`. Each guard costs three private integer coordinates, so the count decreases by $3 \cdot 36 = 108$.

Quantity	Parent	This construction
Independent guards	167	131
Integer variables	888	780
Quadratic residuals	470	434
Collected monomials	10,156	9,628
Exact degree	4	4
Maximum absolute coefficient	2 445 790 974 704	2 445 790 974 704

The variable decrease is 12.16% within this direct quartic family. Other repository constructions have fewer variables at different degrees or coefficient costs. This result makes no claim of global minimality or worldwide priority.

The independent finite checker imports neither the producer nor its elimination module. It verifies source bytes, the staged sign facts, all semantic interfaces, every signed inverse substitution, and the complete collected polynomial. Separate controls require deliberate coefficient, index, guard and restoration corruptions to be rejected; the build is reproduced byte for byte in an isolated directory. These checks authenticate the fixed instance of the proof. The unbounded number-theoretic statements and inherited logical calibration remain the stated mathematical dependencies, and no new end-to-end Lean or object-ZFC proof object is claimed.

5 Verification and reproducible artifacts

The package is designed so that a reader can inspect an equation without first running any source program. The ten-positive equation is given by the full register definitions in this article and by a literal JSON circuit. The 780-integer equation is given by an ordered variable list, all 434 quadratic residuals, and a separately collected sparse expansion. The supplied source files and checks define no hidden computation oracle.

5.1 Ten-positive checks

The coefficient loader checks both gzip and uncompressed hashes, minimal binary lengths, bit lengths, $\beta + 1 = 2^{590069}$, and (2.1). The builder checks that every circuit input and row is live, that no forward reference occurs, and that the literal circuit has exactly 79 multiplications, 33 additions, and 21 subtractions. Squares and multiplication by a fixed scalar each count as one binary multiplication. Fixed integer leaves themselves are free in this cost; the count is not bit complexity.

An independent formula evaluator retains the proof's W abbreviation while the circuit distributes the corresponding product. Exact modular comparison checks their agreement at 100 inputs. Four full univariate restrictions, over two explicitly recorded prime fields, have degree 706 and the proved leading coefficient. The code also checks 150 finite parity/majorant cases and 172 finite canonical-gap cases. The universal upper bound and nonzero top form are mathematical proofs in the degree section, not conclusions drawn from sampled points.

5.2 Direct-quartic checks

The direct checker authenticates the original archive and its selected members, replays the staged sign certificate against literal rows, verifies the exact semantic inputs to the shared frame and each power request, and reconstructs the signed projection. The sparse expansion is independently collected and compared coefficient for coefficient with the delivered polynomial. Its nonzero degree-four terms certify exact degree four. The proof distinguishes these finite algebraic checks from the inherited unbounded computation calibration.

The top-level replay regenerates outputs in temporary directories and compares them with the frozen delivery. Checks are run under ordinary Python and under `python -O`; correctness conditions use explicit exceptions rather than relying on assertions that optimization can disable. A separate AI-assisted review checks the new mathematical proof ordering. Neither the replay nor the review is described as a proof assistant checking the complete consistency theorem.

5.3 Files and exact commands

Path	Purpose
`report/report.tex`, its two section files, and `report/report.pdf`	Editable article and compiled reading copy.
`dyadic10/PROOF.md`	Complete text proof and polynomial recipe.
`dyadic10/circuit.py` and `p10_dyadic_degree706.json`	All 133 arithmetic rows; ten witnesses; fixed coefficient binding.
`dyadic10/coefficients/`	The four literal, unchanged coefficient payloads.
`direct780/equations/`	Literal residual equation, full collected polynomial, sign certificate, signed parent, inverse projection, and exact metrics.
`direct780/PROOF.md`	Complete new delayed-sign and projection proof.
`direct780/upstream/`	Immutable direct parent archive and required extracts.
`provenance/`	Pinned calibration snapshots and source ledger.
`verify.py` and `VERIFICATION.json`	Whole-package replay and receipt.
`MANIFEST.sha256`	Checksums for the final delivered files.

From the extracted package directory, run

```
python3 verify.py
python3 -O verify.py
cd report
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The verification uses the Python standard library. Building the PDF requires a normal LaTeX installation with the packages named in its preamble. The mathematical proof can be read without either runtime.

6 Limitations, interpretation, and further research

6.1 What has and has not been achieved

The concrete outcome is two improvements with explicit domains and complete polynomial data. The first is a smaller degree at a fixed count of ten positive unknowns; the second is a smaller integer quartic within the direct C222 certificate architecture. Both have the required consistency equivalence after composition with the stated inherited calibration theorem. The new proofs are ordinary mathematics and use only finite algebra, divisibility, Pell recurrences, inequalities, and induction that can be expressed in ZFC.

We have not established that either equation is globally simplest. The ten-positive equation does not lower the repository's best known positive variable count from nine. Its 133-operation circuit is longer than the parent's 124-operation circuit. The direct quartic does not beat the variable counts of the native-coefficient quartics; it retains the different direct certificate route and its comparatively modest coefficient height. The parent 407-natural-variable polynomial is not claimed as a new result here.

The native tuple and the C222 source calibration are inherited results, not new independent machine-verification projects. No kernel-checked endpoint for the whole new equivalence is supplied. We also have not established priority in the literature for the broad methods of valuation separation, centering, or recovering quotient signs. The specific new combinations and their full equation data are the bounded research claims of this package.

6.2 Questions suggested by the reductions

1. **Optimize degree without rebuilding root-size barriers.** The dyadic lemma permits deleting the extra strong scale. Can the remaining factor c^2 in $I = ic^2$ be replaced by a smaller expression while retaining the exact auxiliary-index theorem? A proposed change must recover the condition $c \mid m$ used by that theorem; a smaller formula alone is insufficient.

2. **Compress the weighted norm evaluator.** The present circuit uses a general sixteen-fold norm identity. Can its special weighted and centered arguments be evaluated in fewer operations without expanding the degree-706 polynomial? Any saving should be verified as equality of the entire fixed polynomial.

3. **Separate arithmetic and sign inference systematically.** The 780-integer construction recovers the signs of power quotients after identifying their arithmetic values. Which other coordinates have an analogous delayed sign proof? A useful search should operate on proof dependencies, not just syntactic nonnegative monomials.

4. **Optimize the direct system before guarding.** Which quadratic changes of variables preserve a reverse restoration and expose more signs without increasing degree? A simultaneous search over domain proofs and monic eliminations may improve on treating the two optimizations separately.

5. **Choose an explicit simplicity objective.** For example, compare a lexicographic objective of unknown count, degree, and fixed-payload bytes with a description-length objective charging every binary numeral and circuit operation. This would distinguish legitimate improvements from incompatible claims about "simplest."

6. **Close a formalization boundary.** The new dyadic lemma and the power-quotient recurrence are small enough to formalize separately. A full endpoint additionally needs the fixed polynomial semantics, source calibration, and derivation inside the chosen formal presentation of ZFC. These are separate obligations from testing a builder.

A Domain conventions and conversion

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0} = \{1, 2, 3, \ldots\}$. The ten-positive construction can be viewed as a ten-natural equation by substituting $x_j + 1$ for each unknown. This preserves its exact degree and adds at most ten additions to the displayed circuit. It does not make the unknowns unrestricted integers.

If an elementary all-integer conversion is wanted for that same polynomial, the identity

$$u > 0 \quad \longleftrightarrow \quad \exists a, b, c \in \mathbb{Z}\ u = 1 + a^2 + b^2 + c(c+1)$$

follows from the three-squares theorem (equivalently the standard representation of nonnegative integers used in [6]). Applying it independently gives a polynomial in 30 unrestricted integers of degree 1412. This is merely an explicit domain conversion, not a competitive new endpoint, so it is not counted as a third result.

B Attribution and source access

The project snapshot is

`ec80841c07e91be8956ce6ed4af9a9c0b782d3d9`.

The native coefficient files come from package 44; the calibrated source interface is package 61; the immediate positive parent is the in-repository degree-816 candidate. That candidate acknowledges earlier four-root, centering, and residue constructions. This work preserves those credits and adds its own concrete dyadic descaling and degree computation. The direct source is the immutable incoming 407/888 archive. Its shared-frame and direct certificate results are retained as inherited material. The new sign recovery and the resulting 780-variable data are distinguished from that archive.

The package license is MIT No Attribution for new material. Inherited licenses and notices are retained with the material to which they apply. All implementation and review in the present package are AI-assisted; the independent review means a separate analysis task, not an external journal referee or a proof assistant.

References

[1] ProveIt contributors, *Ten positive unknowns, degree 816: a scaled, centered sixteen-fold norm*, candidate and proof at snapshot `ec80841c07e91be8956ce6ed4af9a9c0b782d3d9`. Pinned complete parent proof. Included unchanged in `dyadic10/upstream/PROOF816.md`.

[2] ProveIt contributors, *Congruence carriers*, package 61, Theorem `thm:source`, canonical promises, and Pell lemmas. Pinned LaTeX source.

[3] ProveIt contributors, *Complement-packed native clauses: a smaller complete coefficient tuple*, package 44, native proof and binding. Pinned native proof.

[4] ProveIt incoming research package, *ConZFC_407_888_Quartics*, archive SHA-256 `59f847872298260ee35e9d8e4cd0b52d2d78658ac523981fe060e81f40ca4a59`. Pinned original archive. Included unchanged under `direct780/upstream/`.

[5] J. Bayer, M. David, M. Haßler, Y. Matiyasevich, and D. Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v3, 22 October 2026. `https://arxiv.org/html/2506.20909v3`.

[6] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026. `https://arxiv.org/html/2609.01594v2`.